

Krutik Tikam

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PROFESSIONAL SUMMARY

AI Systems Engineer specializing in agentic workflows, LLM tooling, and high-performance backend architectures. Proven expertise in engineering asynchronous REST APIs, Model Context Protocol (MCP) servers, and local RAG systems using Python, FastAPI, and ChromaDB. Adept at leveraging modern AI development tools including Claude and Cursor to architect scalable, low-latency intelligence platforms and rapidly iterate on complex system designs.

TECHNICAL SKILLS

AI Engineering: Agentic Systems, Model Context Protocol (MCP), Vector Search (ChromaDB), RAG Architectures, Local LLMs (Ollama), SymPy

Systems & Backend: Python, FastAPI (Async), Rust (Tauri), PostgreSQL, SQLite, RESTful APIs, WebSockets

DevOps & Tooling: Docker, Git/GitHub Actions, Cursor, Claude, AWS, Postman

PROFESSIONAL EXPERIENCE

Hudl India Private Limited

Mumbai, India

Wyscout Analyst

Aug 2026 – Present

- Analyze high-frequency sports telemetry and broadcast data to support advanced backend AI pipelines and predictive analytics workflows.
- Leverage proprietary rule frameworks to structure and tag kinematic spatio-temporal features, ensuring high-quality data ingestion for machine learning models.

Talent Corner HR Services Private Limited

Mumbai, India

Software Engineering Intern

Dec 2025 – Feb 2026

- Architected and optimized **PostgreSQL** database schemas to efficiently manage and query a dense relational dataset of **60,000+** detailed professional profiles, significantly minimizing search latency.
- Engineered asynchronous REST APIs using **FastAPI** and Python to power a dynamic authentication system and an automated document generation engine.
- Leveraged AI coding assistants (Cursor/Claude) to accelerate backend pipeline development, deploying via Git and conducting rigorous API load testing with Postman to ensure high-availability system stability.

AI SYSTEMS & BACKEND PROJECTS

AURA HUD: Native Edge Desktop AI Companion | *Tauri, FastAPI, LangChain, SQLite*

- Engineered a cursor-anchored desktop AI assistant utilizing a Rust (Tauri) core to achieve a minimal memory footprint while maintaining sub-3.0s local LLM response latency.
- Architected zero-latency clipboard intelligence using FastAPI and SQLite in Write-Ahead Logging (WAL) mode to instantly parse and action copied code contexts.
- Developed a 100% offline audio streaming pipeline processing voice data over WebSockets into a local speech-to-text engine for secure, high-speed transcription.

Blender MCP Server: LLM Protocol Integration | *Python, Anthropic MCP, TCP Sockets*

- Developed a production-grade Model Context Protocol (MCP) server establishing a direct, real-time interface between multimodal LLMs and live 3D graphics sessions.
- Engineered 24+ tool endpoints allowing agentic systems to autonomously execute Python scripts, inspect scene datablocks, and analyze viewport layouts via JSON.
- Utilized Cursor and Claude for rapid architectural prototyping, seamlessly mapping complex Blender API structures into robust, agent-accessible tool endpoints.

OmniMath-Local: Asynchronous Vector RAG Platform | *FastAPI, ChromaDB, DeepSeek, Python*

- Engineered an enterprise-grade async backend framework serving custom algorithmic validation logic and symbolic math verification workflows.
- Implemented semantic vector search using **ChromaDB** for a local RAG system, indexing over 12,000 technical documents to provide accurate, context-aware mathematical reasoning.
- Optimized system throughput and minimized disk boundaries by executing asynchronous upsert minibatches, maintaining a lean local storage footprint while completely eliminating mathematical hallucinations via dual neural-symbolic validation.

EDUCATION

Nagandas Khandwala College

Mumbai, Maharashtra

B.Sc. Computer Science (Spl. in Artificial Intelligence & Machine Learning)

2023 – 2026

- **Academic Standing:** CGPA: 7.5
- **Relevant Focus:** Data Structures & Algorithms (DSA), Neural Networks, Database Management Systems (DBMS), Distributed Web Architecture.

CERTIFICATIONS

Cloud Engineering: AWS Data Engineering for Generative AI Applications, AWS Connecting Systems and Machines for Industrial Manufacturing, Microsoft Azure Cloud Computing (SDP)

Machine Learning & Data Science: DeepLearning.AI Machine Learning Specialization, TensorFlow Developer Certificate